

# CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

**PRE-MEDICAL : NURTURE COURSE [ PHASE-I & II ]**

**TEST # 12**

**DATE : 23-12-2018**

**Test Type : Minor**

**Test Pattern : NEET**

## INSTRUCTIONS

**Do not open this Test Booklet until you are asked to do so**

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 12

Topics

DATE : 23 - 12 - 2018

## PRE-MEDICAL NURTURE COURSE [ PHASE-I &amp; II ] : [NEET]

PHYSICS : WPE, Circular Motion, SHM, Wave (Upto Wave On A string)

CHEMISTRY : Mole Concept, Redox Reaction, Ionic Equilibrium, s-block, and Hydrogen

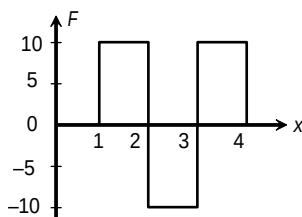
 BIOLOGY : Transport in Plants, Mineral Nutrition and Photosynthesis, Respiration System  
 and Circulatory System

SPACE FOR ROUGH WORK

**HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS**

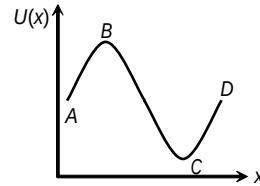
**BEWARE OF NEGATIVE MARKING**

- A gun fires a bullet of mass 50 gm with a velocity of  $30 \text{ m sec}^{-1}$ . Because of this the gun is pushed back with a velocity of  $1 \text{ m sec}^{-1}$ . The mass of the gun is  
(1) 15 kg (2) 30 kg  
(3) 1.5 kg (4) 20 kg
- A force of  $2\hat{i} + 3\hat{j} + 4\hat{k}$  N acts on a body for 4 second, produces a displacement of  $(3\hat{i} + 4\hat{j} + 5\hat{k})\text{m}$ . The power used is  
(1) 9.5 W (2) 7.5 W  
(3) 6.5 W (4) 4.5 W
- An engine pumps up 100 kg of water through a height of 10 m in 5 s. Given that the efficiency of the engine is 60% . If  $g = 10 \text{ ms}^{-2}$ , the power of the engine is  
(1) 3.3 kW (2) 0.33 kW  
(3) 0.033 kW (4) 33 kW
- Masses of two substances are 1 g and 9 g respectively. If their kinetic energies are same, then the ratio of their momentum will be  
(1) 1 : 9 (2) 9 : 1  
(3) 3 : 1 (4) 1 : 3
- Figure shows the  $F$ - $x$  graph. Where  $F$  is the force applied and  $x$  is the distance covered



by the body along a straight line path. Given that  $F$  is in newton and  $x$  in metre, what is the work done ?

- (1) 10 J (2) 20 J  
(3) 30 J (4) 40 J
- The potential energy of a particle varies with distance  $x$  as shown in the graph.



The force acting on the particle is zero at

- (1) C (2) B  
(3) B and C (4) A and D
- A shell is fired from a cannon with velocity  $v \text{ m/sec}$  at an angle  $\theta$  with the horizontal direction. At the highest point in its path it explodes into two pieces of equal mass. One of the pieces retraces its path to the cannon and the speed in  $\text{m/sec}$  of the other piece immediately after the explosion is  
(1)  $3v \cos \theta$  (2)  $3v \cos \theta$   
(3)  $\frac{3}{2}v \cos \theta$  (4)  $\frac{\sqrt{3}}{2}v \cos \theta$
  - A particle of mass  $m$  is moving in a horizontal circle of radius  $r$  under a centripetal force equal to  $-K/r^2$ , where  $K$  is a constant. The total energy of the particle is

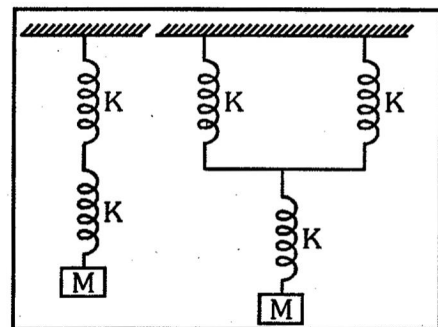
- (1)  $\frac{K}{2r}$  (2)  $-\frac{K}{2r}$   
(3)  $-\frac{K}{r}$  (4)  $\frac{K}{r}$

SPACE FOR ROUGH WORK

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| <p>9. A 50 g bullet moving with velocity 10 m/s strikes a block of mass 950 g at rest and gets embedded in it. The loss in kinetic energy will be</p> <p>(1) 100% (2) 95%</p> <p>(3) 5% (4) 50%</p> <p>10. A uniform chain of length <math>L</math> and mass <math>M</math> is lying on a smooth table and one third of its length is hanging vertically down over the edge of the table. If <math>g</math> is acceleration due to gravity, the work required to pull the hanging part on to the table is</p> <p>(1) <math>MgL</math> (2) <math>MgL/3</math></p> <p>(3) <math>MgL/9</math> (4) <math>MgL/18</math></p> <p>11. If the overbridge is concave instead of being convex, the thrust on the road at the lowest position will be</p> <p>(1) <math>mg + \frac{mv^2}{r}</math> (2) <math>mg - \frac{mv^2}{r}</math></p> <p>(3) <math>\frac{m^2v^2g}{r}</math> (4) <math>\frac{v^2g}{r}</math></p> <p>12. The ratio of angular speeds of minute hand and hour hand of a watch is</p> <p>(1) 1 : 12 (2) 6 : 1</p> <p>(3) 12 : 1 (4) 1 : 6</p> <p>13. An electric fan has blades of length 30 cm as measured from the axis of rotation. If the fan is rotating at 1200 r.p.m. The acceleration of a point on the tip of the blade is about</p> <p>(1) 1600 m/sec<sup>2</sup> (2) 4740 m/sec<sup>2</sup></p> <p>(3) 2370 m/sec<sup>2</sup> (4) 5055 m/sec<sup>2</sup></p> | <p>14. What is the value of linear velocity, if <math>\vec{\omega} = 3\hat{i} - 4\hat{j} + \hat{k}</math> and <math>\vec{r} = 5\hat{i} - 6\hat{j} + 6\hat{k}</math></p> <p>(1) <math>6\hat{i} + 2\hat{j} - 3\hat{k}</math> (2) <math>-18\hat{i} - 13\hat{j} + 2\hat{k}</math></p> <p>(3) <math>4\hat{i} - 13\hat{j} + 6\hat{k}</math> (4) <math>6\hat{i} - 2\hat{j} + 8\hat{k}</math></p> <p>15. A particle of mass <math>M</math> is moving in a horizontal circle of radius <math>R</math> with uniform speed <math>V</math>. When it moves from one point to a diametrically opposite point, its</p> <p>(1) Kinetic energy changes by <math>MV^2/4</math></p> <p>(2) Momentum does not change</p> <p>(3) Momentum changes by <math>2MV</math></p> <p>(4) Kinetic energy changes by <math>MV^2</math></p> <p>16. A particle of mass <math>M</math> moves with constant speed along a circular path of radius <math>r</math> under the action of a force <math>F</math>. Its speed is</p> <p>(1) <math>\sqrt{\frac{rF}{m}}</math> (2) <math>\sqrt{\frac{F}{r}}</math></p> <p>(3) <math>\sqrt{Fmr}</math> (4) <math>\sqrt{\frac{F}{mr}}</math></p> <p>17. An athlete completes one round of a circular track of radius 10 m in 40 sec. The distance covered by him in 2 min 20 sec is</p> <p>(1) 70 m (2) 140 m</p> <p>(3) 110 m (4) 220 m</p> <p>18. In uniform circular motion</p> <p>(1) Both the angular velocity and the angular momentum vary</p> <p>(2) The angular velocity varies but the angular momentum remains constant</p> <p>(3) Both the angular velocity and the angular momentum stay constant</p> <p>(4) The angular momentum varies but the angular velocity remains constant</p> |
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SPACE FOR ROUGH WORK

19. A weightless thread can support tension upto 30 N. A stone of mass 0.5 kg is tied to it and is revolved in a circular path of radius 2 m in a vertical plane. If  $g = 10 \text{ m/s}^2$ , then the maximum angular velocity of the stone will be  
 (1) 5 rad/s  
 (2)  $\sqrt{30}$  rad/s  
 (3)  $\sqrt{60}$  rad/s  
 (4) 10 rad/s
20. A body slides down a frictionless track which ends in a circular loop of diameter  $D$ , then the minimum height  $h$  of the body in term of  $D$  so that it may just complete the loop, is  
 (1)  $h = \frac{5D}{2}$   
 (2)  $h = \frac{5D}{4}$   
 (3)  $h = \frac{3D}{4}$   
 (4)  $h = \frac{D}{4}$
21. A stone tied with a string, is rotated in a vertical circle. The minimum speed with which the string has to be rotated  
 (1) Is independent of the mass of the stone  
 (2) Is independent of the length of the string  
 (3) Decreases with increasing mass of the stone  
 (4) Decreases with increasing in length of the string
22. If seventh harmonic is set up in a string fixed on both ends, number of nodes and antinodes formed are  
 (1) 8, 7  
 (2) 7, 7  
 (3) 8, 9  
 (4) 9, 8
23. A simple pendulum oscillates in a vertical plane. When it passes through the mean position, the tension in the string is 3 times the weight of the pendulum bob. What is the maximum displacement of the pendulum of the string with respect to the vertical  
 (1)  $30^\circ$   
 (2)  $45^\circ$   
 (3)  $60^\circ$   
 (4)  $90^\circ$
24. The acceleration of a particle in SHM at 5 cm from its' mean position is  $20 \text{ cm/sec}^2$ . The value of angular velocity in radian/second will be :  
 (1) 2  
 (2) 4  
 (3) 10  
 (4) 14
25. Some springs are combined in series and parallel arrangement as shown in the figure and a mass  $M$  is suspended from them. The ratio of their frequencies will be :  
 (1) 1 : 1  
 (2) 2 : 1  
 (3)  $\sqrt{3} : 2$   
 (4) 4 : 1



SPACE FOR ROUGH WORK

26. The instantaneous displacement of a simple pendulum oscillator is given by

$x = A \cos\left(\omega t + \frac{\pi}{4}\right)$ . Its speed will be maximum at time :-

- (1)  $\frac{\pi}{4\omega}$  (2)  $\frac{\pi}{2\omega}$  (3)  $\frac{\pi}{\omega}$  (4)  $\frac{2\pi}{\omega}$

27. A block of mass  $m$  rests on a platform. The platform is given up and down SHM with an amplitude  $d$ . What can be the maximum frequency so that the block never leaves the platform?

- (1)  $\sqrt{g/d}$  (2)  $\frac{1}{2\pi}\sqrt{g/d}$

- (3)  $\frac{1}{2\pi}(g/d)$  (4)  $2\pi\sqrt{g/d}$

28. A lift is ascending with acceleration  $g/3$ . What will be the time period of a simple pendulum suspended from its ceiling if its time period in stationary lift is  $T$ ?

- (1)  $\frac{T}{2}$  (2)  $\frac{\sqrt{3}T}{2}$  (3)  $\frac{\sqrt{3}T}{4}$  (4)  $\frac{T}{4}$

29. A simple pendulum performs simple harmonic motion about  $x = 0$  with an amplitude  $a$  and time period  $T$ . The speed of the pendulum at  $x = a/2$  will be :-

- (1)  $\frac{\pi a \sqrt{3}}{T}$  (2)  $\frac{\pi a \sqrt{3}}{2T}$

- (3)  $\frac{\pi a}{T}$  (4)  $\frac{3\pi^2 a}{T}$

30. A horizontal spring is connected to a mass  $M$ . It executes simple harmonic motion. When the mass  $M$  passes through its mean position, an object of mass  $m$  is put on it and the two move together. The ratio of frequencies before and after will be-

- (1)  $\left(1 + \frac{m}{M}\right)^{1/2}$  (2)  $\left(1 + \frac{m}{M}\right)$

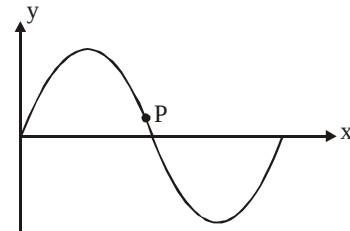
- (3)  $\left(\frac{M}{M+m}\right)^{1/2}$  (4)  $\left(\frac{M}{M+m}\right)$

31. How long after the beginning of motion is the displacement of a harmonically oscillating point equal to one half its amplitude, if the period is 24 sec and initial phase is zero ?

- (1) 12 sec (2) 2 sec

- (3) 4 sec (4) 6 sec

32. A transverse sinusoidal wave moves along a string in the positive  $x$ -direction at a speed of 10 cm/s. The wavelength of the wave is 0.5m and its amplitude is 10 cm. At a particular time  $t$ , the snap-shot of the wave is shown in fig. The velocity of  $P$  when its displacement is 5 cm is :-



- (1)  $\frac{\sqrt{3}\pi}{50} \hat{i}$  m/s (2)  $-\frac{\sqrt{3}\pi}{50} \hat{j}$  m/s

- (3)  $\frac{\sqrt{3}\pi}{50} \hat{j}$  m/s (4)  $-\frac{\sqrt{3}\pi}{50} \hat{i}$  m/s

SPACE FOR ROUGH WORK

33. The speed of a wave on a string is 150 m/s when the tension is 120 N. The percentage increase in the tension in order to raise the wave speed by 20% is:-

(1) 44% (2) 40%  
(3) 20% (4) 10%

34. A particle describes SHM in a straight line about O.



If the time period of the motion is T then its kinetic energy at P be half of its peak value at O, if the time taken by the particle to travel from O to P is

(1)  $\frac{1}{2}T$  (2)  $\frac{1}{4}T$

(3)  $\frac{1}{2\sqrt{2}}T$  (4)  $\frac{1}{8}T$

35. The angular velocity and the amplitude of a simple pendulum are  $\omega$  and A respectively. At a displacement x from the mean position if its kinetic energy is T and potential energy is U, then the ratio of T to U is:

(1)  $\frac{x^2\omega^2}{(A^2 - x^2\omega^2)}$

(2)  $\frac{x^2}{(A^2 - x^2)}$

(3)  $\frac{A^2 - x^2\omega^2}{(x^2 - x^2\omega^2)}$

(4)  $\frac{A^2 - x^2}{x^2}$

36. The speed of transverse waves in a stretched string is 700 cm/s. If the string is 2 m long, the frequency with which is resonates in fundamental mode is :-

(1)  $(7/2)$  Hz  
(2)  $(7/4)$  Hz  
(3) (14) Hz  
(4)  $(2/7)$  Hz

37. A body executes S.H.M. with an amplitude A. At what displacement from the mean position, is the potential energy of the body one-fourth of its total energy?

(1)  $\frac{A}{4}$

(2)  $\frac{A}{2}$

(3)  $\frac{3A}{4}$

(4) Some other fraction of A

38. The time period of oscillations of a simple pendulum is 1 minute. If its length is increased by 44%, then its new time period of oscillation will be:-

(1) 96 s (2) 58 s  
(3) 82 s (4) 72 s

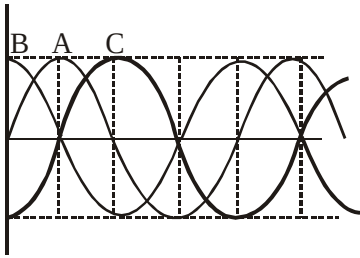
39. The equation of stationary wave in a stretched string is given by

$y = 5 \sin(\pi x/3) \cos(40\pi t)$  where x and y are in cm and t in sec. The separation between two adjacent nodes is :-

(1) 1.5 cm (2) 3 cm  
(3) 6 cm (4) 4 cm

SPACE FOR ROUGH WORK

40. Three progressive wave A, B and C are shown in fig. with respect to wave A :-



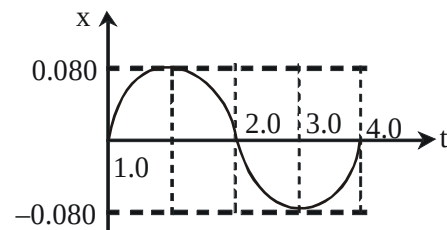
- (1) the wave C lags behind in phase by  $\pi/2$  and B leads by  $\pi/2$   
 (2) the wave C leads in phase by  $\pi$  and B lags behind by  $\pi$   
 (3) the wave C leads in phase by  $\pi/2$  and B lags behind by  $\pi/2$   
 (4) the wave C lags behind in phase by  $\pi$  and B leads by  $\pi$
41. The equation of motion of a particle executing simple harmonic motion is  $a + 16\pi^2x = 0$ . In this equation,  $a$  is the linear acceleration in  $\text{m/s}^2$  of the particle at a displacement  $x$  in metre. Find the time period.
- (1) 0.50 (2) 0.15  
 (3) 0.155 (4) 0.25
42. A wave equation which gives the displacement along y-direction is given by;  
 $y = 10^{-4} \sin (60t + 2x)$   
 where  $x$  and  $y$  are in m and  $t$  in s. This represents a wave :-  
 (i) travelling with velocity of 30 m/s in the negative x-direction  
 (ii) of wavelength ( $\pi$ ) m  
 (iii) of frequency ( $30/\pi$ ) Hz  
 (iv) of amplitude  $10^{-4}$  m travelling along negative x-axis

- (1) (iii, iv)  
 (2) (ii, iv)  
 (3) (ii, iii)  
 (4) (i, ii, iii, iv)

43. A particle of mass  $m$  is executing S.H.M. If amplitude is  $a$  and frequency  $n$ , the value of its force constant will be :

- (1)  $mn^2$  (2)  $4mn^2a^2$   
 (3)  $ma^2$  (4)  $4\pi^2mn^2$

44. An object of mass 0.8 kg is attached to one end of a spring and the system is set into simple harmonic motion. The displacement  $x$  of the object as a function of time is shown in the figure. The magnitude of the object's acceleration at  $t = 1.0$  is



- (1) zero  
 (2)  $1.57 \text{ m/s}^2$   
 (3)  $0.197 \text{ m/s}^2$   
 (4)  $0.157 \text{ m/s}^2$
45. A particle executes SHM of type  $x = a \sin \omega t$ . It takes time  $t_1$  from  $x = 0$  to  $x = \frac{a}{2}$  and  $t_2$  from  $x = \frac{a}{2}$  to  $x = a$ . The ratio of  $t_1 : t_2$  will be :-
- (1) 1 : 1 (2) 1 : 2  
 (3) 1 : 3 (4) 2 : 1

SPACE FOR ROUGH WORK

46. Match list – I (compounds) with list – II (Oxidation state of nitrogen) and select the correct answer using the codes given below the lists –

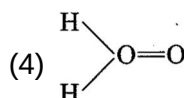
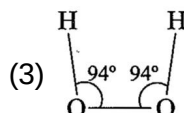
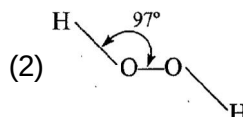
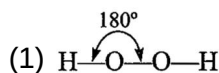
| List – I                   | List – II  |
|----------------------------|------------|
| (A) $\text{NaN}_3$         | (a) +5     |
| (B) $\text{N}_2\text{H}_4$ | (b) +2     |
| (C) NO                     | (c) $-1/3$ |
| (D) $\text{N}_2\text{O}_5$ | (d) –2     |

Codes :–

| A     | B | C | D |
|-------|---|---|---|
| (1) c | d | b | a |
| (2) d | c | b | a |
| (3) c | d | a | b |
| (4) d | c | a | b |

47. The formula which represents the simple ratio of atoms in a compound is called –
- molecular formula
  - structural formula
  - empirical formula
  - rational formula
48. The oxidation state of tungsten in  $\text{Na}_2\text{W}_4\text{O}_{13} \cdot 10\text{H}_2\text{O}$  is –
- +7
  - +6
  - +4
  - +4.5
49. Metal hydrides are ionic, covalent or molecular in nature. Among LiH, NaH, KH, RbH, CsH, the correct order of increasing ionic character is:
- $\text{LiH} > \text{NaH} > \text{CsH} > \text{KH} > \text{RbH}$
  - $\text{LiH} < \text{NaH} < \text{KH} < \text{RbH} < \text{CsH}$
  - $\text{RbH} > \text{CsH} > \text{NaH} > \text{KH} > \text{LiH}$
  - $\text{NaH} > \text{CsH} > \text{RbH} > \text{LiH} > \text{KH}$

50. Which of the following is a true structure of  $\text{H}_2\text{O}_2$ ?



51. Equivalent weight of  $\text{Mn}^{3+}$  in the following reaction is (Mn = 55)



- 27.5
- 55
- 110
- 165

52. The reaction  $\text{H}_2\text{S} + \text{H}_2\text{O}_2 \longrightarrow 2\text{H}_2\text{O} + \text{S}$  shows :–

- oxidising nature of  $\text{H}_2\text{O}_2$
- reducing action of  $\text{H}_2\text{O}_2$
- alkaline nature of  $\text{H}_2\text{O}_2$
- acidic nature of  $\text{H}_2\text{O}_2$

53. *Para* and *ortho* hydrogen differ in:–

- atomic number
- atomic mass
- spins of proton
- number of neutrons

SPACE FOR ROUGH WORK

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| <p>54. The reducing power of a metal depends on various factors. Suggest which makes Li, the strongest reducing agent in aqueous solution?</p> <p>(1) Sublimation enthalpy<br/>           (2) Ionisation enthalpy<br/>           (3) Hydration energy<br/>           (4) Electron gain enthalpy</p> <p>55. 200 ml <math>\frac{N}{10}</math> HCl is mixed with 150 ml <math>\frac{N}{5}</math> NaOH and resulting solution is made up to 1 litre. The pH of resulting solution will be</p> <p>(1) 2      (2) 4      (3) 10      (4) 12</p> <p>56. For alkali metals, which one of the following trends is incorrect?</p> <p>(1) Hydration energy : <math>\text{Li}^+ &gt; \text{Na}^+ &gt; \text{K}^+ &gt; \text{Rb}^+</math><br/>           (2) Ionisation energy : <math>\text{Li} &gt; \text{Na} &gt; \text{K} &gt; \text{Rb}</math><br/>           (3) Density : <math>\text{Li} &lt; \text{Na} &lt; \text{K} &lt; \text{Rb}</math><br/>           (4) Atomic size : <math>\text{Li} &lt; \text{Na} &lt; \text{K} &lt; \text{Rb}</math></p> <p>57. Lithium does not resemble with other alkali metals in the following properties. Which is incorrect?</p> <p>(1) <math>\text{Li}_2\text{CO}_3</math> decomposes on heating while other alkali metal carbonates are stable towards heat<br/>           (2) Lithium directly reacts with nitrogen while other alkali metals do not react<br/>           (3) Lithium like other alkali elements does not combine directly with carbon<br/>           (4) LiCl is soluble in pyridine while chlorides of other alkali metals are insoluble in pyridine</p> | <p>58. Lithium is the only alkali metal which is not placed in kerosene but wrapped in paraffin wax, because:</p> <p>(1) it reacts with kerosene<br/>           (2) it floats to the surface of kerosene because of low density<br/>           (3) it does not react with air and <math>\text{H}_2\text{O}</math><br/>           (4) none of the above</p> <p>59. When sodium is dissolved in liquid ammonia, a solution of blue colour is obtained. The colour of the solution is due to :-</p> <p>(1) ammoniated electron<br/>           (2) sodium ion<br/>           (3) sodium amide<br/>           (4) ammoniated sodium ion</p> <p>60. A chemical A' is used for the preparation of washing soda to recover ammonia. When <math>\text{CO}_2</math> is bubbled through an aqueous solution of A', the solution turns milky. It is used in white washing due to disinfectant nature. What is chemical formula of A'?</p> <p>(1) <math>\text{Ca}(\text{HCO}_3)_2</math><br/>           (2) CaO<br/>           (3) <math>\text{Ca}(\text{OH})_2</math><br/>           (4) <math>\text{CaCO}_3</math></p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

SPACE FOR ROUGH WORK

61. Which sequence of reactions shows chemical relation between sodium and its compounds?
- (1)  $\text{Na} + \text{O}_2 \longrightarrow \text{Na}_2\text{O} \xrightarrow{\text{HCl}} \text{NaCl}$   
 $\xrightarrow{\text{CO}_2} \text{Na}_2\text{CO}_3 \xrightarrow{\text{Heat}} \text{Na}$
- (2)  $\text{Na} + \text{O}_2 \longrightarrow \text{Na}_2\text{O} \xrightarrow{\text{H}_2\text{O}} \text{NaOH}$   
 $\xrightarrow{\text{CO}_2} \text{Na}_2\text{CO}_3 \xrightarrow{\text{Heat}} \text{Na}$
- (3)  $\text{Na} + \text{H}_2\text{O} \longrightarrow \text{NaOH} \xrightarrow{\text{HCl}} \text{NaCl}$   
 $\xrightarrow{\text{CO}_2} \text{Na}_2\text{CO}_3 \xrightarrow{\text{Heat}} \text{Na}$
- (4)  $\text{Na} + \text{H}_2\text{O} \longrightarrow \text{NaOH} \xrightarrow{\text{CO}_2} \text{Na}_2\text{CO}_3$   
 $\xrightarrow{\text{HCl}} \text{NaCl} \xrightarrow[\text{(Molten)}]{\text{Electrolysis}} \text{Na}$
62. Correct order of stability of group IIA metal carbonates is :-
- (1)  $\text{BaCO}_3 > \text{SrCO}_3 > \text{CaCO}_3 > \text{MgCO}_3$   
 (2)  $\text{MgCO}_3 > \text{CaCO}_3 > \text{SrCO}_3 > \text{BaCO}_3$   
 (3)  $\text{SrCO}_3 > \text{BaCO}_3 > \text{CaCO}_3 > \text{MgCO}_3$   
 (4)  $\text{CaCO}_3 > \text{MgCO}_3 > \text{BaCO}_3 > \text{SrCO}_3$
63. A solid compound 'X' on heating gives  $\text{CO}_2$  gas and a residue. The residue mixed with water forms 'P'. On passing an excess of  $\text{CO}_2$  through 'P' in water, a clear solution 'Z' is obtained. On boiling 'Z' compound 'X' is reformed. The compound 'X' is :-
- (1)  $\text{Ca}(\text{HCO}_3)_2$   
 (2)  $\text{CaCO}_3$   
 (3)  $\text{Na}_2\text{CO}_3$   
 (4)  $\text{K}_2\text{CO}_3$
64. Which of the following halides of calcium is insoluble in water?
- (1)  $\text{CaCl}_2$  (2)  $\text{CaI}_2$   
 (3)  $\text{CaF}_2$  (4)  $\text{CaBr}_2$
65. 50 ml of  $\text{HCl}$  ( $\text{pH} = 1$ ) is mixed with 100 ml of  $\text{HCl}$  ( $\text{pH} = 2$ ) then the  $\text{pH}$  of resulting solution will be ( $\log 2 = 0.3$ )
- (1) 1.7 (2) 1.6  
 (3) 1.4 (4) 1.2
66. Heavy water is :-
- (1) water containing  $\text{CaSO}_4$ ,  $\text{MgSO}_4$   
 (2) water containing  $\text{CaCl}_2$   
 (3)  $\text{D}_2\text{O}$   
 (4) water containing  $\text{CaSO}_3$
67. Permanent hardness of water is due to the presence of :-
- (1) bicarbonates of sodium and potassium  
 (2) chlorides and sulphates of sodium and potassium  
 (3) chlorides and sulphates of calcium and magnesium  
 (4) bicarbonates of calcium and magnesium
68. Which of the following contains the largest number of atoms -
- (1) 11g of  $\text{CO}_2$   
 (2) 4g of  $\text{H}_2$   
 (3) 5g of  $\text{NH}_3$   
 (4) 8g of  $\text{SO}_2$
69. A compound was found to contain 5.37% nitrogen by mass. What is the minimum molecular weight of compound-
- (1) 26.07 (2) 2.607  
 (3) 260.7 (4) None

SPACE FOR ROUGH WORK

70. Eq. wt. of  $\text{MnO}_4^-$  in acidic, basic, neutral medium are in the ratio of -

  - 3 : 5 : 15
  - 5 : 3 : 1
  - 5 : 1 : 3
  - 3 : 15 : 5

71. Why does  $\text{H}^+$  ion always get associated with other atoms or molecules?

  - Ionisation enthalpy of hydrogen resembles that of alkali metals
  - Its reactivity is similar to halogens
  - It resembles both alkali metals and halogens
  - Due to very small size it cannot exist free

72. Two oxides of Metal contain 27.6% and 30% oxygen respectively. If the formula of first oxide is  $\text{M}_3\text{O}_4$  then formula of second oxide is -

  - MO
  - $\text{M}_2\text{O}$
  - $\text{M}_2\text{O}_3$
  - $\text{MO}_2$

73. The vapour density of gas A is four times that of B. If molecular mass of B is M, then molecular mass of A is -

  - M
  - 4M
  - $\frac{M}{4}$
  - 2M

74. Oxidation numbers of two Cl atoms in bleaching powder,  $\text{CaOCl}_2$  is -

  - 1, -1
  - +1, -1
  - +1, +1
  - 0, -1

75. Which of the following statements is incorrect?

  - Syn. gas is used in Bosch's process for the production of  $\text{H}_2$
  - Hydrogen adsorbed by palladium is termed occluded hydrogen
  - The isotopes deuterium and tritium are radioactive in nature
  - Hydrogen is used in the hydrogenation of oils in presence of nickel as a catalyst

76. In which property listed below hydrogen does not resemble alkali metals?

  - Nature of oxide
  - Reducing nature
  - Tendency to form cation
  - Presence of one electron in valence shell

77. In the reaction,

$$3\text{Br}_2 + 6\text{NaOH} \rightarrow \text{NaBrO}_3 + 5\text{NaBr} + 3\text{H}_2\text{O}$$

which element loses as well as gains electrons in the above reaction -

  - Na
  - Br
  - O
  - H

78. Oxygen shows oxidation state of -1 in the compound -

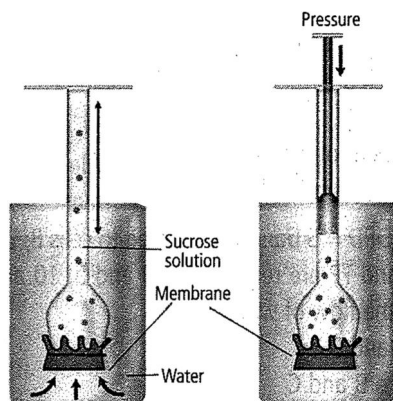
  - $\text{NO}_2$
  - $\text{MnO}_2$
  - $\text{PbO}_2$
  - $\text{Na}_2\text{O}_2$

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79. In the reaction,  
 $2\text{Al}_{(s)} + 6\text{HCl}_{(aq)} \rightarrow 2\text{Al}^{3+}_{(aq)} + 6\text{Cl}^{-}_{(aq)} + 3\text{H}_{2(g)}$   
 (1) 6L  $\text{HCl}_{(aq)}$  is consumed for every 3L  $\text{H}_{2(g)}$  produced  
 (2) 33.6 L  $\text{H}_{2(g)}$  is produced regardless of temperature and pressure for every mole Al that reacts  
 (3) 67.2 L  $\text{H}_{2(g)}$  at STP is produced for every mole Al that reacts  
 (4) 11.2 L  $\text{H}_{2(g)}$  at STP is produced for every mole  $\text{HCl}_{(aq)}$  consumed
80. Equivalent weight of  $\text{FeC}_2\text{O}_4$  in the change,  
 $\text{FeC}_2\text{O}_4 \rightarrow \text{Fe}^{3+} + \text{CO}_2$  is  
 (1) M/3 (2) M/6  
 (3) M/2 (4) M/1
81. Which of the following compounds is used for water softening?  
 (1)  $\text{Ca}_3(\text{PO}_4)_2$  (2) NaCl  
 (3)  $\text{Na}_6\text{P}_6\text{O}_{18}$  (4)  $\text{Na}_2\text{HPO}_4$
82. A mixture containing 100 gm  $\text{H}_2$  and 100 gm  $\text{O}_2$  is ignited so that water is formed according to the reaction,  $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ ; How much water will be formed -  
 (1) 113 gm (2) 50 gm  
 (3) 25 gm (4) 200 gm
83. The stoichiometric coefficients of  $\text{MnO}_4^-$ ,  $\text{Pb}^{2+}$  and  $\text{H}_2\text{O}$  in the balanced chemical equation of the reaction  
 $\text{Mn}^{2+} + \text{PbO}_2 + \text{H}^+ \rightarrow \text{MnO}_4^- + \text{Pb}^{2+} + \text{H}_2\text{O}$ ,  
 respectively are -  
 (1) 2, 5, 2 (2) 1, 4, 2  
 (3) 1, 2, 6 (4) 1, 2, 3
84. The solubility of  $\text{Ag}_2\text{CrO}_4$  is  $2.49 \times 10^{-2}$  gm per litre. The solubility product of  $\text{Ag}_2\text{CrO}_4$  is (Cr = 52; Ag = 108)  
 (1)  $5.6 \times 10^{-10}$  (2)  $1.6875 \times 10^{-12}$   
 (3)  $6.2 \times 10^{-5}$  (4)  $1.86 \times 10^{-8}$
85. A buffer solution is maximum effective when pH is  
 (1)  $\text{pK}_a$  (2)  $\text{pK}_a + 1$   
 (3)  $\text{pK}_a - 1$  (4)  $\frac{\text{pK}_a}{\text{pK}_a + 1}$
86. Which is the correct order of pH of following aqueous solution?  
 (1)  $\text{NH}_4\text{Cl} < \text{CH}_3\text{COONH}_4 < \text{CH}_3\text{COONa}$   
 (2)  $\text{CH}_3\text{COONa} < \text{CH}_3\text{COONH}_4 < \text{NH}_4\text{Cl}$   
 (3)  $\text{CH}_3\text{COONH}_4 < \text{CH}_3\text{COONa} < \text{NH}_4\text{Cl}$   
 (4)  $\text{NH}_4\text{Cl} < \text{CH}_3\text{COONa} < \text{CH}_3\text{COONH}_4$
87. At  $90^\circ\text{C}$  pure water has  $[\text{H}_3\text{O}^+] = 10^{-6}\text{mol/litre}$ . The value at  $K_w$  at  $90^\circ\text{C}$  is  
 (1)  $10^{-6}$  (2)  $10^{-8}$   
 (3)  $10^{-12}$  (4)  $10^{-14}$
88. An acidic buffer solution can be prepared by mixing solution of  
 (1) Sodium acetate and acetic acid  
 (2) Ammonia and ammonium hydroxide  
 (3) Sulphuric acid and sodium sulphate  
 (4) Sodium chloride and sodium hydroxide
89. Which of the metal carbonates is most stable?  
 (1)  $\text{MgCO}_3$  (2)  $\text{CaCO}_3$   
 (3)  $\text{SrCO}_3$  (4)  $\text{BaCO}_3$
90. Which compound is highly covalent?  
 (1) LiF (2) LiCl  
 (3) LiBr (4) LiI

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91. Refer to the given figure and select the correct option regarding this.



- (1) Water will diffuse across the membrane to raise the level of solution in the funnel.  
(2) Pressure can be applied to prevent osmosis  
(3) Sucrose solution is an osmotically active solution which can absorb water.  
(4) All of these
92. Match column I with column II and select the correct option from the codes given below.

| Column-I |                   | Column-II |                      |
|----------|-------------------|-----------|----------------------|
| A.       | Photospectrometer | (i)       | Root pressure        |
| B.       | Porometer         | (ii)      | Transpiration        |
| C.       | Manometer         | (iii)     | Osmotic pressure     |
| D.       | Osmometer         | (iv)      | Absorption spectrum  |
| E.       | Potometer         | (v)       | stomatal conductance |

- (1) A-(iv), B-(iii), C-(v), D-(ii), E-(i)  
(2) A-(iv), B-(v), C-(i), D-(iii), E-(ii)  
(3) A-(iv), B-(iii), C-(i), D-(ii), E-(v)  
(4) A-(v), B-(iv), C-(iii), D-(ii), E-(i)
93. Choose the correct option w.r.t Mycorrhiza
- A. absorption of water  
B. absorption of inorganic and few nitrogenous organic solute

- C. mutualistic relationship  
D. gaseous exchange.  
E. translocation of solute  
F. protection from pathogens  
G. exclusively in higher plants  
H. biofertilizer  
I. most of the species of glomus genus form symbiotic relationship with plants root.

- J. essential for germination of pinus seed

- (1) A, C, F, H, J (2) A, B, C, G, H  
(3) A, C, E, I, J (4) A, C, F, I, J

94. The values of osmotic potential ( $\pi$ ) and pressure potential ( $\rho$ ) of cells A, B, C and D are given below.

| Cell | $\pi$ | $\rho$ |
|------|-------|--------|
| A    | -1.0  | 0.5    |
| B    | -0.6  | 0.3    |
| C    | -1.2  | 0.6    |
| D    | -0.8  | 0.4    |

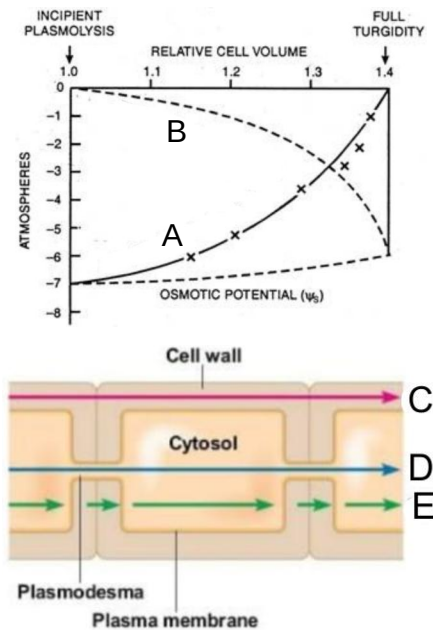
Identify the option, which shows correct sequence of the path of movement of water.

- (1) A  $\rightarrow$  B  $\rightarrow$  C  $\rightarrow$  D (2) B  $\rightarrow$  D  $\rightarrow$  A  $\rightarrow$  C  
(3) B  $\rightarrow$  C  $\rightarrow$  D  $\rightarrow$  A (4) D  $\rightarrow$  A  $\rightarrow$  B  $\rightarrow$  C

95. Read the following statements and choose the correct option.

- I. Translocation mainly involves xylem and phloem tissue.  
II. Water channels are formed of 4 different types of proteins.  
III. Uphill transport of materials require energy input.  
IV. Imbibition is reversible phenomenon  
V. Pure water has highest water potential. Its value is zero.  
VI. Diffusion is most obvious in gases and solids  
VII. High pH favours stomatal opening  
(1) I, II, IV, V  
(2) II, III, IV, VI, VII  
(3) I, II, IV, V, VII  
(4) I, IV, V, VII

96. Refer to the given figures. Identify the labelled parts (A-E) and select the correct option.



- (1) A - Water potential;  
B - Pressure potential;  
C - Apoplast pathway;  
D - Vacuolar pathway;  
E - Symplast pathway
- (2) A - Pressure potential;  
B - Water potential;  
C - Symplast pathway;  
D - Transmembrane pathway;  
E - Apoplast pathway
- (3) A - Water potential;  
B - Pressure potential;  
C - Apoplast pathway;  
D - Symplast pathway;  
E - Transmembrane pathway
- (4) A - Turgor pressure  
B - Water potential;  
C - Apoplast pathway;  
D - Symplast pathway;  
E - Vacuolar pathway

97. How many of the following events are occurring at source for phloem?

- A - Synthesis of sugar  
B - Active unloading of sugar  
C - Conversion of glucose to sucrose  
D - High osmotic pressure  
E - Low hydrostatic pressure

- (1) 1 (2) 2  
(3) 3 (4) 4

98. Girdling (or ringing) experiments prove that :-

- (1) Xylem tissues are located inner to phloem  
(2) Pith is located inner to xylem  
(3) Inorganic minerals are translocated via xylem elements  
(4) Organic content is mainly translocated through phloem

99. Due to transpiration pull, flow rate of water may reach upto :-

- (1) 15 cm/hr  
(2) 50 cm/hr  
(3) 15 m/hr  
(4) 50 m/hr

100. How much of the absorbed water is lost during transpiration?

- (1) < 1% (2) 5-10 %  
(3) 40-50 % (4) > 99 %

101. Match the following :-

| Column-A |                              | Column-B |    |
|----------|------------------------------|----------|----|
| A.       | Photolysis of water          | 1.       | Zn |
| B.       | Component of nitrogenase     | 2.       | Mn |
| C.       | Component of ferredoxin      | 3.       | S  |
| D.       | Biosynthesis of auxin        | 4.       | Mo |
| E.       | Synthesis of mitotic spindle | 5.       | Ca |

- (1) A-2, B-4, C-1, D-5, E-3  
(2) A-2, B-4, C-3, D-1, E-5  
(3) A-3, B-1, C-2, D-4, E-5  
(4) A-1, B-3, C-2, D-5, E-4

102. Various steps of mechanism of nodule formation in legumes are given below. Arrange them in correct order.  
 A - Formation of bacteroids  
 B - Rhizobium multiply around root hairs  
 C - Infection thread release bacteria into the cells of inner cortex  
 D - Curling of root hair  
 E - Formation of nodules  
 F - Establishment of direct vascular connection with host  
 (1) B → D → C → E → A → F  
 (2) D → B → E → A → F → C  
 (3) D → B → A → C → F → E  
 (4) B → D → C → A → E → F
103. During transamination, which amino acid commonly donates the  $\text{NH}_2^-$  group?  
 (1) Glutamic acid  
 (2) Glycine  
 (3) Aspartic acid  
 (4) Valine
104. Which of the following statements are correct about mineral absorption in plants?  
 (1) In the initial phase, ions are taken up into the outer space of cells, the apoplast. It is a passive process  
 (2) In the final phase, ions are taken slowly into the inner space, the symplast of cells and it is an active process  
 (3) Passive movement of the ions into the apoplast occurs through ion channels, transmembrane proteins, which act as selective pores  
 (4) All of the above
105. The deficiencies of micronutrients not only affect the growth of the plants, but also its vital functions, such as photosynthetic and mitochondrial electron flow. Among the list given below, which group of three elements will mostly affect, both photosynthetic and mitochondrial electron transport?  
 (1) Co, Ni and Mo  
 (2) Cu, Mn and Fe  
 (3) Mn, Co and Ca  
 (4) Ca, K and Na
106. The minerals can be detected even at the concentration as low as  
 (1)  $10^{-5}$  g/mL  
 (2)  $10^{-7}$  g/mL  
 (3)  $10^{-8}$  g/mL  
 (4) None of the above
107. Read the following statements and select the correct ones.  
 (i) The co-ordinated activities of the legume and *Rhizobium* bacteria depend on chemical interactions between the symbiotic partners,  
 (ii) Leguminous roots secrete chemical attractants that attract *Rhizobium* bacteria living nearby.  
 (iii) N, P and K usually do not get deficient in soil due to their low plant requirement.  
 (iv) In hydroponics technique, plant can be grown in sandy soil with nutrient solution.  
 (v)  $\text{Mg}^{+2}$  is essential for photolysis of water  
 (1) (i), (ii) and (v)  
 (2) (i), (ii), and (iv)  
 (3) (i), (ii) and (iii)  
 (4) (i), (ii), (iv) and (v)
108. Assertion : Amides contain more nitrogen than amino acids.  
 Reason : They are transported to other parts of plants via phloem.  
 (1) If both Assertion & Reason are true & the reason is a correct explanation of the Assertion.  
 (2) If both Assertion & Reason are true but reason is not a correct explanation of the Assertion.  
 (3) If Assertion is true but the Reason is false  
 (4) If both Assertion & Reason are false

109. Iron :

- (A) Required in larger amounts in comparison to other micronutrients
- (B) It activates enzyme like hexokinase and nitrogenase
- (C) It is essential for chlorophyll synthesis
- (1) Only two options are correct
- (2) Only one option is incorrect
- (3) Only two options are incorrect
- (4) All are correct

110. How many of the given elements are required by plants excess of 10 m mole  $\text{Kg}^{-1}$  dry mater?

Carbon, Phosphorus, Manganese, Copper, Boron, sulphur, Potassium, Calcium, Chlorine, Iron, Magnesium

- (1) 8      (2) 6      (3) 4      (4) 7

111. Mo is not required for :-

- (1) Nitrogen metabolism
- (2) Nitrogenase activity
- (3) Nitrite reductase activity
- (4) Nitrate reductase activity

112. Which of the following statement is correct for sulphur and calcium?

- (1) Mobile elements
- (2) Deficiency symptoms first appear in older leaves
- (3) Relatively immobile and are not transported out of the mature organs
- (4) Micronutrients

113. Spinach leaves are green because :-

- (1) Chlorophyll absorbs green light
- (2) Excited electrons emit green light at they fall back to their resting state.
- (3) Green light is not used in photosynthesis
- (4) Chlorophyll reflects green light

114. Which of the following scientist showed that it is only the green part of the plants that would release oxygen?

- (1) Joseph Priestley
- (2) Jan Ingenhousz
- (3) T. W. Engelmann
- (4) C. van Neil

115. Action spectrum of photosynthesis resembles roughly the

- (1) Absorption spectra of chlorophyll a
- (2) Absorption spectra of chlorophyll b
- (3) Absorption spectra of chlorophyll c
- (4) Absorption spectra of chlorophyll a and b

116. Isotopes employed to study photosynthesis are

- (1)  $^{11}\text{C}$  and  $^{32}\text{P}$
- (2)  $^{15}\text{C}$  and  $^{32}\text{P}$
- (3)  $^{16}\text{C}$  and  $^{15}\text{N}$
- (4)  $^{14}\text{C}$  and  $^{18}\text{O}$ .

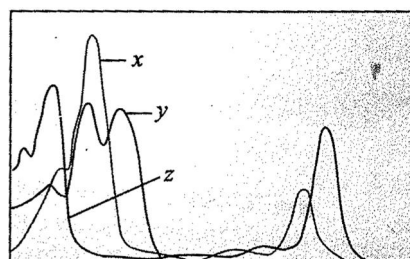
117. At carbon atom III, chlorophyll a has

- (1) Methyl group
- (2) Aldehyde group
- (3) Carboxyl group
- (4) Magnesium

118. Solarisation is

- (1) Formation of chlorophyll
- (2) Destruction of chlorophyll
- (3) Utilisation of sunlight
- (4) Effect of solar light

119. Recognise the figure and find out the correct matching.



- (1) x-chlorophyll a, y-chlorophyll b, z-carotenoids
- (2) z-chlorophyll a, y-chlorophyll b, x-carotenoids
- (3) z-chlorophyll a, x-chlorophyll b, y-carotenoids
- (4) y-chlorophyll a, x-chlorophyll b, z-carotenoids

120. The evidence that during photosynthesis oxygen comes from water

- (1) Photosynthetic bacteria employ  $H_2S$  and  $CO_2$  to form carbohydrates, water and sulphur
- (2) Isolated illuminated chloroplasts release oxygen if provided with potassium ferrocyanide
- (3) Isotopic  $^{18}O$  provided as  $H_2^{18}O$  appears as  $^{18}O_2$  liberated in photosynthesis
- (4) All the above

121. Visible light =  $x \text{ \AA}$

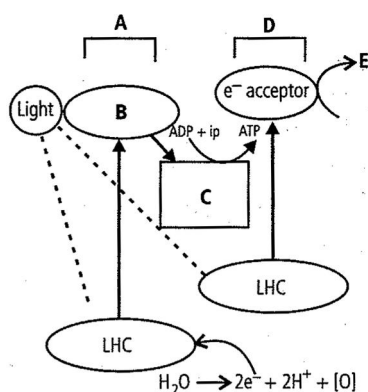
PAR =  $y \text{ \AA}$

- (1)  $x = 4000 - 7000$   
 $y = 3900 - 7600$
- (2)  $x = 3900 - 7600$   
 $y = 4000 - 7000$
- (3)  $x = 4000 - 7000$   
 $y = 4000 - 7000$
- (4)  $x = 3900 - 7600$   
 $y = 3900 - 7600$

122. Movement of electrons from electron acceptor through the electron transport system having cytochromes during non-cyclic photo-phosphorylation can be defined as :-

- (1) Uphill transport
- (2) Downhill transport
- (3) Z-Scheme
- (4) Active transport

123. Identify A, B, C, D and E in the given flow chart showing Z-scheme of light reaction.



|     | A               | B              | C                      | D               | E        |
|-----|-----------------|----------------|------------------------|-----------------|----------|
| (1) | $P_{700}$       | $H^+$ acceptor | $e^-$ acceptor         | $P_{680}$       | $NADP^+$ |
| (2) | Photo-system I  | $e^-$ acceptor | $e^-$ transport system | Photo-system II | $NADPH$  |
| (3) | Photo-system II | $H^+$ acceptor | $e^-$ acceptor         | $P_{700}$       | $NADPH$  |
| (4) | Photo-system II | $e^-$ acceptor | $e^-$ transport system | Photo-system I  | $NADPH$  |

124. Study the following statements.

- (i) Red light falling in the range of wavelength 660-760 nm is the most effective for photosynthesis.
- (ii) Green light falling in the range of wavelength 500-580 nm is the least effective for photosynthesis.
- (iii) Chl *a*, chl *b*, carotenes and xanthophylls are soluble in organic solvents.
- (iv) Phycobilins (phycocyanin, allophycocyanin and phycoerythrin) are soluble in water.

Which of the above statements is/are incorrect ?

- (1) (ii) and (iii)
- (2) (iii) and (iv)
- (3) (i) only
- (4) None of these

125. Which one of the following equations suggests that  $O_2$  released during photosynthesis comes from water?

- (1)  $6CO_2^{18} + 12H_2O \rightarrow 6O_2^{18} + C_6H_{12}O_6 + 6H_2O^{18}$
- (2)  $6CO_2 + 12H_2O^{18} \rightarrow 6O_2 + C_6H_{12}O_6 + 6H_2O^{18}$
- (3)  $6CO_2^{18} + 12H_2O \rightarrow 6O_2^{18} + C_6H_{12}O_6 + 6H_2O$
- (4)  $6CO_2 + 12H_2O^{18} \rightarrow 6O_2^{18} + C_6H_{12}O_6 + 6H_2O$

126. Julius von Sachs in 1854 stated that

- (1) green parts of plant release  $O_2$
- (2) green parts of plant consume  $CO_2$
- (3) glucose in green parts is stored as starch
- (4) plants release phlogisticated air

127. In intense light, chloroplast align themselves in the mesophyll cell in such a way that their flat surfaces are

- (1) antiparallel to the cell wall
- (2) perpendicular to the cell wall
- (3) parallel to the cell wall
- (4) middle in the cell

128. The main photosynthetic pigments in the higher plants are

- (1) chlorophyll-a and chlorophyll-c
- (2) chlorophyll-a and chlorophyll-b
- (3) chlorophyll-b and chlorophyll-a
- (4) chlorophyll-b and chlorophyll-c

129. The movement of electrons in ETC in light reaction is

- (1) uphill in terms of redox reaction
- (2) downhill in terms of redox reaction
- (3) uphill in terms of oxidation
- (4) Both (1) and (2)

130. Light Harvesting Complex (LHC) is

- (1) one molecule of chlorophyll-a
- (2) very few molecules of chlorophyll-a
- (3) hundreds of pigment molecules bound to proteins
- (4) chlorophyll-a + chlorophyll-c + protein + DNA

131. Match the given pigments in column A with their colour given in Column B :-

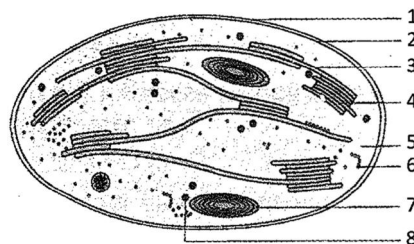
| Column-A |               | Column-B |               |
|----------|---------------|----------|---------------|
| A.       | Chlorophyll-a | 1.       | Blue green    |
| B.       | Chlorophyll-b | 2.       | Yellow        |
| C.       | Xanthophylls  | 3.       | Yellow orange |
| D.       | Carotenoids   | 4.       | Yellow green  |

- (1) A-4, B-1, C-2, D-3
- (2) A-1, B-4, C-2, D-3
- (3) A-4, B-2, C-3, D-1
- (4) A-1, B-3, C-2, D-4.

132. Carotenoids show their absorption peaks in :-

- (1) Blue region
- (2) Red region
- (3) Blue and red region
- (4) Red and Yellow region

133. Section of chloroplast is given in the diagram. Which of the following is correctly matched with its function.



|     | Cyclic Photo-phosphorylation | Non-cyclic photo-phosphorylation | Dark Reaction | Storage of starch |
|-----|------------------------------|----------------------------------|---------------|-------------------|
| (1) | 3                            | 4                                | 2             | 7                 |
| (2) | 4                            | 3                                | 2             | 6                 |
| (3) | 3                            | 4                                | 5             | 7                 |
| (4) | 1                            | 2                                | 5             | 8                 |

134. What is the difference between the mitochondrial and the photosynthetic electron transport chain?

- (1) The final electron acceptor in photosynthesis is  $\text{NADP}^+$  while in respiration it is  $\text{O}_2$
- (2) The mitochondrial electron transport chain is more powerful than the photosynthetic electron transport chain.
- (3) In photosynthesis, the electron transport chain creates glucose while in respiration it creates ATP.
- (4) In photosynthesis, the electron transport chain is located in the outer membrane while in respiration it is located in the inner membrane.

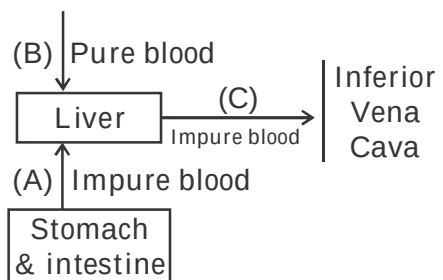
135. Photosynthesis involves the movement of electrons. Which of the following sequence correctly represents the flow of electrons during photosynthesis?

- (1)  $\text{CO}_2 \rightarrow \text{ATP} \rightarrow \text{G3P}$
- (2)  $\text{H}_2\text{O} \rightarrow \text{NADP}^+ \rightarrow \text{G3P}$
- (3)  $\text{CO}_2 \rightarrow \text{NADPH} \rightarrow \text{O}_2$
- (4)  $\text{H}_2\text{O} \rightarrow \text{G3P} \rightarrow \text{O}_2$

136. In a study of four blood vessels A, B, C & D in man, vessel 'A' carried pure blood away from heart. Vessel 'B' carried impure blood towards heart. Vessel 'C' carried pure blood into the heart musculature while vessel 'D' carried impure blood away from heart. The correct names of the four vessels can be :

- (1) Coronary artery-A, Pulmonary vein-B, Pulmonary Artery-C, SVC-D
- (2) Pulmonary vein-A, Pulmonary Artery-B, Coronary vein-C, Coronary Artery-D
- (3) Systemic Aorta-A, Vena cava-B, Coronary artery-C, Pulmonary vein-D
- (4) Systemic Aorta-A, Vena cava-B, Coronary Artery-C, Pulmonary Artery-D

137. Please observe the diagram below



Here vessel A, B and C are -

- (1) A - Hepatic vein  
B - Portal vein  
C - Hepatic artery
- (2) A - Portal vein  
B - Hepatic artery  
C - Hepatic vein
- (3) A - Hepatic artery  
B - Hepatic vein  
C - Portal vein
- (4) A - Portal vein  
B - Hepatic vein  
C - Hepatic artery

138. Which statement is wrong about partial pressure of oxygen.

- (1)  $P_{O_2}$ , more in atmospheric air than alveoli
- (2)  $P_{O_2}$  more in alveoli than tissue
- (3)  $P_{O_2}$  more in oxygenated blood than deoxygenated blood
- (4)  $P_{O_2}$  more in deoxygenated blood than oxygenated blood

139. Dead space air is :-

- (1) The amount of air remaining in the alveoli
- (2) The amount of air left behind in lungs at the end of deep expiration
- (3) The amount of air taken in and out
- (4) The air left in the bronchial tree

140. Consider the following statements and select the option which includes all the correct ones :-

- (a) Partial pressures of  $O_2$  and  $CO_2$  in Alveolar air are 104 mmHg and 40 mm Hg
- (b) Partial pressures of  $O_2$  and  $CO_2$  in deoxygenated blood are 40 mmHg and 45 mm Hg respectively
- (c) Partial pressures of  $O_2$  and  $CO_2$  in deoxygenated blood are 95 mmHg and 40 mm Hg
- (d) Partial pressures of  $O_2$  and  $CO_2$  in Atmospheric air are 40 mmHg and 46 mm Hg

Options :-

- (1) Statements a, b and c are correct
  - (2) Statements a, b, c and d are correct
  - (3) Statements a, b are correct
  - (4) Statements c and d are correct
141. In a human cardiac cycle which of the following statement is false :-
- (1) The time taken for one cardiac cycle decreases in Tachycardia
  - (2) The time interval for joint Diastole in heart is about 0.4 seconds
  - (3) The time interval for joint systole in heart is about 0.4 second
  - (4) Left ventricle takes 0.5 seconds to complete one diastole

142. Which Respiratory capacity is correctly matched with its formula & value.

| Respiratory Capacities           | Formula        | Value   |
|----------------------------------|----------------|---------|
| (1) Functional Residual Capacity | ERV + TV       | 2300 ml |
| (2) Vital Capacity               | ERV + TV + IRV | 4600 ml |
| (3) Total Capacity               | RV + IRV + ERV | 5800 ml |
| (4) Inspiratory Capacity         | TV + IRV       | 1600 ml |

143. What will be the  $P_{O_2}$  and  $P_{CO_2}$  in the atmospheric air compared to those in the alveolar air ?

- (1)  $P_{O_2}$  lesser,  $P_{CO_2}$  higher
- (2)  $P_{O_2}$  higher,  $P_{CO_2}$  lesser
- (3)  $P_{O_2}$  higher,  $P_{CO_2}$  higher
- (4)  $P_{O_2}$  lesser,  $P_{CO_2}$  lesser

144. In which the following respiratory tubule dead space volume is not found ?

- (1) Primary bronchi
- (2) Alveolar duct
- (3) Segmental bronchi
- (4) Intra Pulmonary bronchioles

145. Total volume of air, a person can expire after a normal inspiration -

- (1) TV + IRV
- (2) Only TV
- (3) TV + ERV
- (4) TV + FRV + IRV

146. The conducting part of respiratory tract does not help in :-

- (1) Transport the atmospheric air to the alveoli.
- (2) Clear air from foreign particles
- (3) Humidify and bring the air to body temperature
- (4) Diffusion of  $O_2$  and  $CO_2$  between blood and air

147. Total volume of air accommodated in the lungs at the end of a forced inspiration :

- (1) ERV + RV
- (2) TV + ERV
- (3) TV + ERV + IRV
- (4) TV + ERV + IRV + RV

148. Which statements is incorrect ?

- (1) Volume of air remaining in the lungs after a normal breathing is 2300 ml
- (2) Inspiration can occur if the pressure within the lungs is less than atmospheric pressure
- (3) A healthy man can inspire or expire approximately 6000 to 8000 ml of air per minute
- (4) When the  $P_{CO_2}$  is high and  $P_{O_2}$  is low as in the alveoli, dissociation of  $CO_2$  from carbamino-haemoglobin takes place

149. Which point is incorrect about transport of gases?

- (1) Nearly 20-25 percent of  $CO_2$  is transported by RBC
- (2) About 7 percent of  $CO_2$  is carried in a dissolved state through plasma.
- (3) About 97 percent of  $CO_2$  is transported by plasma of blood.
- (4) About 3 percent of  $O_2$  is carried in a dissolved state through the plasma.

150. Consider the following four statements (a-d) and select the option which includes all the correct :-

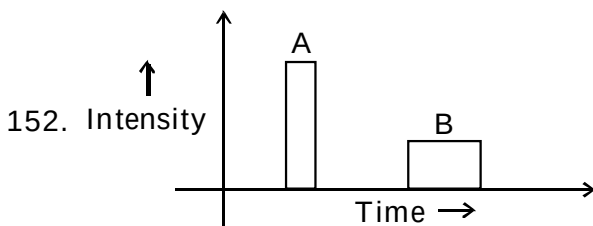
- (a) Coronary Artery disease. (CAD) often referred to as Atherosclerosis
- (b) Heart failure means when the heart muscle is suddenly damaged by an inadequate blood supply
- (c) High blood pressure leads to heart diseases and also affects vital organs like brain and kidney
- (d) Angina occurs due to conditions that affect the blood flow

Options :-

- (1) Statements (b), (c) and (d)
- (2) Statements (a), (b)
- (3) Statements (b), (d)
- (4) Statement (a), (c) and (d)

151. Given below is a table showing various blood groups and donor compatibility. Identify the incorrect option :-

| Blood Group | Antigens on RBCs | Antibodies in Plasma | Donor's Group |
|-------------|------------------|----------------------|---------------|
| (1) A       | A                | anti-B               | A, O          |
| (2) B       | B                | anti-B               | B, O          |
| (3) AB      | A, B             | anti-A, B            | AB, A, B, O   |
| (4) O       | nil              | anti-A, B            | O             |

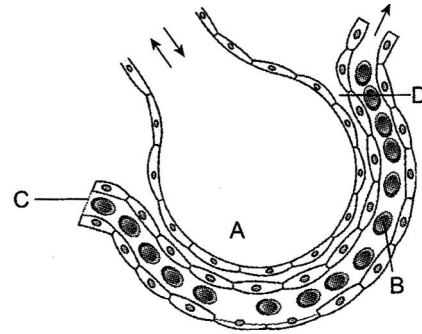


- (A) Sharp sound with high pitch  
 (B) Appears at the beginning of joint diastole  
 (C) Appears at the beginning of ventricular systole  
 (D) Appears at the beginning of ventricular diastole.  
 (E) Appears 0.3 seconds past after atrial systole  
 (F) Caused due to the closure of semilunar valves  
 (G) Caused due to the closure of cuspid valves

Which of the following statements are true for sound 'A' ?

- (1) A, B, D, E and F  
 (2) A, B, D and F  
 (3) A, D, E and F  
 (4) A, B, C, E and G

153. The figure given below shows a small part of human lung where exchange of gases takes place. In which one of the options given below, one part A, B, C or D is correctly identified along with its function,



Options :

- (1) B : Red blood cell—transport of  $\text{CO}_2$  mainly  
 (2) C : Arterial capillary—passes oxygen to tissues  
 (3) A : Alveolar cavity—main site of exchange of respiratory gases  
 (4) D : Capillary wall—exchange of  $\text{O}_2$  and  $\text{CO}_2$  takes place here
154. Name the chronic respiratory disorder caused mainly by cigarette smoking.  
 (1) Emphysema  
 (2) Asthma  
 (3) Respiratory acidosis  
 (4) Respiratory alkalosis
155. Reduction in pH of blood will  
 (1) Reduce the rate of heart beat  
 (2) Reduce the blood supply to the brain  
 (3) Decrease the affinity of haemoglobin with oxygen  
 (4) Release bicarbonate ions by the liver
156. Lungs are made up of air-filled sacs, the alveoli. They do not collapse even after forceful expiration, because of  
 (1) Inspiratory reserve Volume  
 (2) Tidal Volume  
 (3) Expiratory Reserve Volume  
 (4) Residual Volume
157. Which of the following vitamin is essential for blood clotting?  
 (1) Vitamin D  
 (2) Vitamin E  
 (3) Vitamin K  
 (4) Vitamin  $\text{B}_{10}$

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| <p>158. Blood clotting can be prevented by adding oxalates or citrates which form insoluble complex with</p> <ol style="list-style-type: none"> <li>(1) Fibrin</li> <li>(2) Thrombin</li> <li>(3) Stable factor</li> <li>(4) Calcium</li> </ol> <p>159. Atrioventricular node (AV node) is present in</p> <ol style="list-style-type: none"> <li>(1) Right upper corner of right atrium</li> <li>(2) Lower left corner of right atrium</li> <li>(3) Right ventricle</li> <li>(4) Left atrium</li> </ol> <p>160. Human heart is myogenic because</p> <ol style="list-style-type: none"> <li>(1) Sympathetic nerves can increase heart rate</li> <li>(2) Impulse for heart beat is generated in SA node</li> <li>(3) Impulse for heart beat is generated outside the heart</li> <li>(4) Myocardium is present in heart wall</li> </ol> <p>161. In the foetal heart, ductus arteriosus connects</p> <ol style="list-style-type: none"> <li>(1) Pulmonary artery with pulmonary vein</li> <li>(2) Pulmonary artery with systemic aorta</li> <li>(3) Right atrium and left atrium</li> <li>(4) Pulmonary vein with systemic aorta</li> </ol> <p>162. The exchange of materials between blood and interstitial fluid takes place through</p> <ol style="list-style-type: none"> <li>(1) Veins</li> <li>(2) Arteries</li> <li>(3) Arterioles</li> <li>(4) Capillaries</li> </ol> <p>163. Slowing down of heart rate is called</p> <ol style="list-style-type: none"> <li>(1) Tachycardia</li> <li>(2) Bradycardia</li> <li>(3) Bradypnoea</li> <li>(4) Dextrocardia</li> </ol> <p>164. Heart rate is accelerated by</p> <ol style="list-style-type: none"> <li>(1) Parasympathetic nerves and acetylcholine</li> <li>(2) Sympathetic nerves and epinephrine</li> <li>(3) Sympathetic nerves and acetylcholine</li> <li>(4) Parasympathetic nerves and epinephrine</li> </ol> | <p>165. If vagus nerve is stimulates, the heart rate</p> <ol style="list-style-type: none"> <li>(1) Increases</li> <li>(2) Decreases</li> <li>(3) Remains unchanged</li> <li>(4) First increases, then decreases</li> </ol> <p>166. Ventricles are related to</p> <ol style="list-style-type: none"> <li>(1) Heart</li> <li>(2) Brain</li> <li>(3) Both (1) and (2)</li> <li>(4) Kidneys</li> </ol> <p>167. Maximum surface area of circulating system is seen in:</p> <ol style="list-style-type: none"> <li>(1) Heart</li> <li>(2) Veins</li> <li>(3) Arterioles</li> <li>(4) Capillaries</li> </ol> <p>168. Which of the following is absent in humans?</p> <ol style="list-style-type: none"> <li>(1) Hepatic vein</li> <li>(2) Hepatic portal vein</li> <li>(3) Renal portal vein</li> <li>(4) Hypophysial portal vein</li> </ol> <p>169. Hardening and loss of elasticity of the arteries is called</p> <ol style="list-style-type: none"> <li>(1) Atherosclerosis</li> <li>(2) Arteriosclerosis</li> <li>(3) Stroke</li> <li>(4) Heart failure</li> </ol> <p>170. Inability of heart to pump sufficient blood is called</p> <ol style="list-style-type: none"> <li>(1) Heart block</li> <li>(2) Heart failure</li> <li>(3) Heart attack</li> <li>(4) Angina pectoris</li> </ol> <p>171. Which of the following is correct regarding ECG?</p> <ol style="list-style-type: none"> <li>(1) By counting the number of QRS complexes that occur in a given time period, the heart beat rate of a person can be determined</li> <li>(2) P-wave represents the electrical excitation of the atria that leads to contraction of both atria</li> <li>(3) T-wave represents the returning of ventricles from excited to the normal state</li> <li>(4) All are correct</li> </ol> |
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| <p>172. Cancerous proliferation of WBCs is called</p> <ol style="list-style-type: none"> <li>(1) Leucocytosis</li> <li>(2) Leukemia</li> <li>(3) Leucopenia</li> <li>(4) Polycythemia</li> </ol> <p>173. Platelets are formed due to breakdown of large cells called:</p> <ol style="list-style-type: none"> <li>(1) Erythroblast</li> <li>(2) Myeloblast</li> <li>(3) Leucoblast</li> <li>(4) Megakaryocyte</li> </ol> <p>174. The role of platelets is</p> <ol style="list-style-type: none"> <li>(1) Transport oxygen</li> <li>(2) Provide immunity</li> <li>(3) Secrete structural proteins</li> <li>(4) Blood clotting</li> </ol> <p>175. Erythroblastosis foetalis can occur if the blood group of father and mother is respectively</p> <ol style="list-style-type: none"> <li>(1) Rh -ve and Rh +ve</li> <li>(2) Rh +ve and Rh +ve</li> <li>(3) Rh +ve and Rh -ve</li> <li>(4) Rh -ve and Rh -ve</li> </ol> <p>176. Rhesus antigen is present</p> <ol style="list-style-type: none"> <li>(1) On the surface of WBCs</li> <li>(2) In plasma</li> <li>(3) On the surface of RBCs</li> <li>(4) On the surface of platelets</li> </ol> | <p>177. A patient with unknown blood group needed blood transfusion immediately after getting injured in a road accident. His one doctor friend at once offers his blood. What was the blood group of the donor?</p> <ol style="list-style-type: none"> <li>(1) Blood group AB</li> <li>(2) Blood group A</li> <li>(3) Blood group B</li> <li>(4) Blood group O.</li> </ol> <p>178. Maximum urea levels are present in</p> <ol style="list-style-type: none"> <li>(1) Hepatic portal vein</li> <li>(2) Hepatic vein</li> <li>(3) Renal vein</li> <li>(4) Hepatic artery</li> </ol> <p>179. Hypophyseal portal veins carry blood from hypothalamus to</p> <ol style="list-style-type: none"> <li>(1) Anterior pituitary</li> <li>(2) Posterior pituitary</li> <li>(3) Thyroid</li> <li>(4) Thymus</li> </ol> <p>180. An artery has</p> <ol style="list-style-type: none"> <li>(1) Thin wall and blood moves at high pressure</li> <li>(2) Thick wall and blood moves at high pressure</li> <li>(3) Thin wall and blood moves at low pressure</li> <li>(4) Thick wall and blood moves at low pressure</li> </ol> |
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